TRANSCRIPT

Title	Module Overview
Module	COM7021 Data Visualisation
Lesson	1
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Timestamp	Content
00:00:21.175	Hello everybody and welcome the data visualisation module as part of the Data Science Master's program. This introductory video is designed to help you get an overall picture of what is included in this exciting and informative Master's module. Plus, briefly explain the constituent parts of the assignment brief.
00:00:42.485	Throughout the entire lesson, we will be using the coding language Python to create all data visualisations and subsequent dashboard. The reason why we are using Python is because this particular language is widely used and its inherent libraries within many data analytics environments. Therefore, by using Python, it will get you well prepared for life after your education and into industry.
00:01:17.765	The module itself will be delivered over a ten-week period, with nine associated face-to-face workshops. There is, however, no face-to-face workshop in week ten as the tenth week has been purely designed to give online assignment support from the module leader.
00:01:38.185	Within week one, we will be discussing what data visualisation is and how this has been developed in both society and industry and give the difference between visualisation and computer graphics. Also, in week one, we will demonstrate some basic visualisations in matplotlib and Seabourne, both graphically based libraries in Python.

00:02:05.005	Week two, we shall take a look at fully understanding your data – a vital part of any data investigation – and then progress onto what are the ideal charts to use within a given data environment. Also, within lesson two, we will demonstrate how to create cloud maps and Gantt charts.
00:02:28.975	Week three will all be focused upon storytelling, visualisation of the variables, perception, and finally culminating with storytelling examples.
00:02:41.795	Week four is concerned with spatial visualisations, beginning with one-dimension through to three-dimension visualisations, completing the lesson by giving theory and examples on dynamic visualisations. In other words, creating data visualisations so a data analysis can be conducted in real time.
00:03:08.755	Week five sees the development from week four to geospatial data visualisations. This is where we look at data from a much wider perspective and associate it with points on a map.
00:03:24.335	Week six, we'll demonstrate what temporal visualisations are through different time variant plots, from both a single and multi-variable perspective.
00:03:36.785	Week seven will be focused on interactive visualisations, in other words, line and bubble charts. Plus, the implementation of adding sliders into the visualisation to fully engage in the interactive element.
00:03:53.965	Week eight will see us dive into the world of appropriate colour application to charts and creating dashboards in Python using the '.dash' library.

00:04:05.035	Finally, in week nine we shall look at the common errors and pitfalls that new data visualisation creators experience and how to overcome them. As you can no doubt determine, a comprehensive yet exciting module full of data visualisation concepts.
00:04:27.555	The assignment brief, to briefly discuss, encompasses all that you will have learned in the entirety of the module. There are two tasks associated with the assignment – a visual investigation on a prescribed data set and an in-depth report on your findings through appropriate storytelling and visualisation, all of which is to be conducted in Python.
00:04:54.195	You will submit a single Word document with screenshotted evidence of the processes undertaken, along with full reasoning and justifications for your analysis. The Python code will also be submitted in the appendix at the end of your submission so that it can be checked and verified it is performing correctly.
00:05:18.325	All that's left for me to say is that the academic team and I are very excited in bringing you into the exciting world of data visualisations using Python. See you in the first workshop and all the very best of luck.